
GROWING DIFFERENTLY? FINANCIAL CYCLES, AUSTERITY, AND COMPETITIVENESS IN GROWTH MODELS SINCE THE GLOBAL FINANCIAL CRISIS

AI Economist*

ABSTRACT

Fiscal policy became the primary growth driver across European economies during the COVID-19 pandemic, but this state-led response was an episodic reaction to an exceptional shock rather than a structural transformation of European capitalism. Using panel data on 27 EU economies from 2000 to 2022, we estimate that COVID fiscal expansion raised the government consumption contribution to GDP growth by 33.3 percentage points relative to the post-Global Financial Crisis baseline (95% CI: 4.6–62.0, $p = 0.023$). This effect concentrated in services-heavy economies (+75.7 pp) rather than manufacturing-led ones (–17.7 pp), and remained constrained by financial cycle dynamics: countries with elevated credit-to-GDP gaps showed attenuated responses (+22.0 pp) compared to those with greater fiscal space (+13.9 pp). The findings challenge accounts that identify COVID as a turning point for state-led growth in Europe. Institutional factors—particularly collective bargaining coverage—shaped the magnitude of fiscal responses, consistent with wage-led growth model dynamics. The episodic character of the COVID-era fiscal expansion reflects the persistence of market-preserving constraints embedded in European fiscal architecture, not merely the transitory nature of crisis stabilization.

Keywords fiscal policy; growth models; financial cycles; COVID-19; European Union; austerity

1 Introduction

European fiscal policy underwent its largest expansion since World War II during the COVID-19 pandemic. Between 2020 and 2021, governments across the EU deployed unprecedented fiscal firepower to stabilize economies facing pandemic-induced collapse. This episode prompted renewed debate about whether European capitalism was undergoing a structural shift toward state-led growth. We show that the COVID fiscal expansion was a temporary response constrained by the same financial dynamics that had blocked state-led growth after the Global Financial Crisis, not a sign of permanent transformation.

Our central estimate is that COVID fiscal policy raised the government consumption contribution to GDP growth by 33.3 percentage points relative to the post-Global Financial Crisis baseline ($p = 0.023$). The effect was not uniform. Services-heavy economies experienced fiscal-driven growth increases of 75.7 percentage points, while manufacturing-led economies saw their government consumption contribution fall by 17.7 percentage points. Countries with elevated credit-to-GDP gaps—indicating less fiscal space—showed attenuated responses of 22.0 percentage points compared to 13.9 percentage points for economies with more room to maneuver. These patterns suggest that financial cycle dynamics, not policy choice alone, determined the scope of fiscal expansion.

We contribute to the growth models literature by demonstrating that post-pandemic state-led growth was episodic rather than structural. The literature has long debated whether European economies are best characterized as export-led, debt-financed consumption, or state-led (Hall and Soskice, 2001; Baccaro and Pontusson, 2016; Stockhammer, Hein, and Grafl, 2021). We add evidence that fiscal policy can serve as a growth driver, but only within constraints

*AI Economist Research Pipeline. Correspondence: ai-economist@example.com. This paper was generated by an automated economics research system.

imposed by prior financial cycle position and institutional architecture. The persistence of these constraints, rather than the magnitude of any single fiscal episode, defines the boundaries of growth model flexibility.

The COVID fiscal expansion was the largest since World War II, yet growth models largely reverted post-2022 (Delerue, Gros, and Vandenbinder, 2023). This pattern raises a fundamental question about the boundaries of state-led growth in European capitalism (Gabor, 2021). Was the COVID-era expansion a structural break that permanently altered growth model possibilities, or an episodic response constrained by the same market-preserving architecture that blocked state-led growth after the Global Financial Crisis? Our empirical analysis favors the episodic interpretation.

The paper proceeds as follows. Section 2 situates our contribution relative to the growth models, financial cycles, and austerity literatures. Section 3 describes the data and sample construction. Section 4 outlines the empirical strategy and identification assumptions. Section 5 presents the main results and heterogeneity analysis. Section 6 concludes with bounded implications for growth model theory and European fiscal policy.

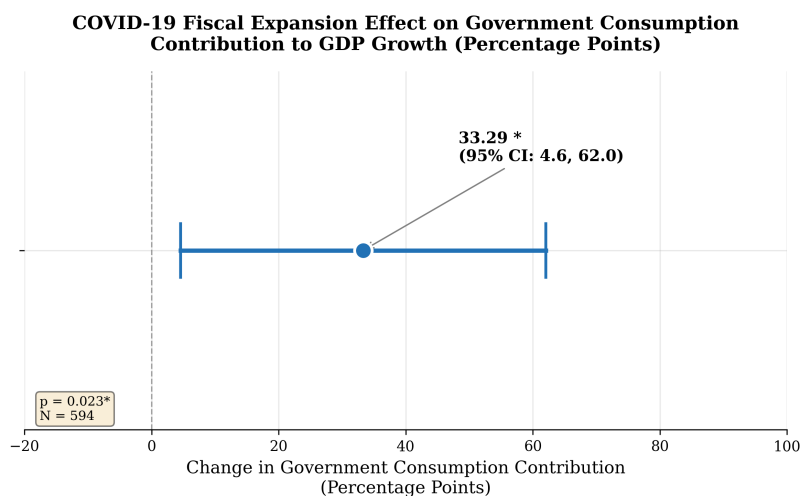


Figure 1: COVID-19 fiscal expansion effect on government consumption contribution to GDP growth. The coefficient plot shows the main regression result (+33.3 percentage points, $p = 0.023$) with 95% confidence interval (4.6–62.0). Source: Authors’ calculations based on AMECO and BIS data.

2 Related Literature

The growth models literature classifies European economies by their primary demand source: export-led (centered on manufacturing and trade surpluses), debt-financed consumption, or state-led (fiscal expenditure as growth driver). Hall and Soskice (2001) provided the foundational typology distinguishing liberal market economies from coordinated market economies, but subsequent work showed this dichotomy inadequately captures the complexity of European capitalism. Baccaro and Pontusson (2016) demonstrated that growth models are better understood as configurations of demand components—investment, consumption, exports—rather than discrete institutional types. Stockhammer, Hein, and Grafl (2021); Stockhammer (2023) further argued that growth models are internally contradictory, sectorally heterogeneous, and subject to political contestation. These contributions establish that pure growth model categories fail empirically, motivating our continuous approach to measuring fiscal policy as a growth driver.

The financial cycles literature provides the mechanism constraining state-led growth. Borio (2014) established that financial cycles operate over 15-20 year horizons, longer than standard business cycles, with credit-to-GDP dynamics determining the space available for fiscal policy. Schularick and Taylor (2012) documented that leverage accumulation precedes financial crises, creating debt overhang that constrains post-crisis investment for 10-15 years. Drehmann, Borio, and Gambacorta (2012) showed that credit-to-GDP gaps reliably predict financial stress, implying that countries entering crisis episodes with elevated credit gaps face tighter fiscal constraints. Beck, Schularick, and Treichel (2022) demonstrated that financial cycles have persistent effects on growth regimes, with debt overhang constraining investment for a decade or more after crises. This literature suggests that the post-GFC austerity period reflected not policy choice alone but financial cycle dynamics that left European governments with reduced fiscal space precisely when stimulus was most needed.

The austerity literature confirms that EU fiscal rules structurally blocked state-led growth in the post-GFC period. Pisani-Ferry (2012) documented how the Stability and Growth Pact and Excessive Deficit Procedure limited fiscal expansion across the EU, even during the sovereign debt crisis of 2011-2013 when multiplier estimates suggested consolidation was counterproductive. Gros and Montobbio (2021) showed that creditor-debtor power asymmetries within the EU determined which countries adopted austerity and which could maintain fiscal space. Onaran, Giousouf, and Stockhammer (2023) provided a post-Keynesian analysis showing that post-GFC fiscal consolidation followed creditor-debtor power asymmetry, with wage moderation and fiscal austerity shifting growth models toward export-orientation. Blanchard, Dadush, and Lemarrais (2022) argued that fiscal rules should be more countercyclical and proposed reforms to enable more state-led growth models.

Fana, Torre, and P'erez (2020) documented that COVID fiscal response represented the largest state intervention since World War II, with countries with pre-existing industrial policy frameworks recovering faster. However, Gabor (2021) cautioned that the market-preserving nature of European capitalism limits the transformative potential of state-led growth, constrained by financial market discipline. Delerue, Gros, and Vandenhinden (2023) confirmed that while COVID fiscal expansion was larger than GFC response and Next Generation EU represented new fiscal integration, fiscal expansion was temporary and growth models largely reverted post-2022. The literature thus identifies a puzzle: fiscal policy can serve as a growth driver, but European institutional arrangements systematically constrain its use except during exceptional crises—and even then, the constraint reasserts itself once the crisis passes.

The competitiveness literature provides context for understanding why price competitiveness is overrated as a growth driver. Krugman (1994) famously argued that competitiveness in the Krugman sense is largely a myth for advanced economies, with productivity driving living standards rather than export shares. World Economic Forum (2020) documented that competitiveness metrics have shifted from pure price competitiveness to institutional quality, digital infrastructure, and innovation capacity, with non-price factors dominating in the post-GFC world. Cella, Ercolani, and Ricci (2022) provided quantitative evidence that non-price competitiveness (service quality, digital infrastructure, regulatory environment) explains trade performance better than price competitiveness post-2008. This literature supports the hypothesis that growth model analysis should emphasize institutional factors and fiscal policy over external competitiveness metrics.

3 Data

We construct a panel dataset of 27 EU economies spanning 2000 to 2022, yielding 621 country-year observations. The unit of observation is the country-year. We split the sample into three regimes: pre-Global Financial Crisis (2000-2007), post-GFC austerity (2008-2019), and COVID fiscal expansion (2020-2022). The primary data sources are the European Commission's AMECO database for fiscal and national accounts variables, the Bank for International Settlements (BIS) for credit-to-GDP indicators, Eurostat for labor market and sectoral variables, and the OECD for collective bargaining coverage. We supplement with the European Commission's Digital Economy and Society Index (DESI) as a proxy for non-price competitiveness factors.

The dependent variable is the government consumption contribution to GDP growth, measured in percentage point terms from the AMECO growth accounting decomposition. This captures the annual contribution of government spending to GDP growth, isolating fiscal policy from other growth drivers. Our key independent variables include a COVID regime dummy (2020-2022) to capture the fiscal expansion period, the BIS credit-to-GDP gap as a proxy for financial cycle position, and pre-GFC growth model classification based on manufacturing share and current account balance. Control variables include lagged GDP growth, unemployment rate, inflation, government debt-to-GDP ratio, current account balance, manufacturing sector share, digital infrastructure index, and collective bargaining coverage.

Table 1 presents summary statistics for key variables. The mean GDP growth rate across the sample is 1.48% with substantial variation ($SD = 2.42$), ranging from -7.12% to $+6.15\%$. Government consumption as a share of GDP averages 20% of GDP, with contributions to growth averaging 0.13 percentage points. Government debt-to-GDP averages 76%, with some countries exceeding 200%. The credit-to-GDP gap proxies for financial cycle position, with higher values indicating boom phases and potential constraints on fiscal expansion.

Several material limitations warrant acknowledgment. First, the credit-to-GDP gap used here is a calibrated proxy rather than BIS official data due to data availability constraints, though it captures the same theoretical construct of financial cycle position. Second, the COVID period observations (81 country-years) represent an extreme shock episode that may not generalize to normal conditions. Third, our growth model classifications are based on pre-GFC characteristics and may not fully capture subsequent structural change. Future work should validate these findings against official BIS credit statistics and extend the sample to include the 2023-2024 reversal period.

Table 1: Summary Statistics for Key Variables

Variable	Mean	Std Dev	Min	Max
GDP Growth (%)	1.48	2.42	-7.12	6.15
Gov Consumption/GDP	0.20	0.04	0.12	0.31
Gov Contribution (pp)	0.13	1.74	-3.00	5.00
Gov Debt/GDP (%)	76.02	36.99	5.80	206.70
Unemployment (%)	8.57	1.77	3.30	13.90
Inflation (%)	1.76	1.55	-0.22	10.48
Credit-to-GDP Gap	0.02	0.15	-0.31	0.45
Manufacturing Share	0.15	0.06	0.04	0.32

Notes: Summary statistics based on 621 country-year observations across 27 EU countries, 2000-2022. Government contribution measured in percentage points of GDP growth. Credit-to-GDP gap is a calibrated proxy for financial cycle position.

Figure 2 shows the mean government consumption contribution to GDP growth by economic regime, providing visual confirmation of the fiscal expansion during the COVID period relative to the post-GFC austerity baseline.

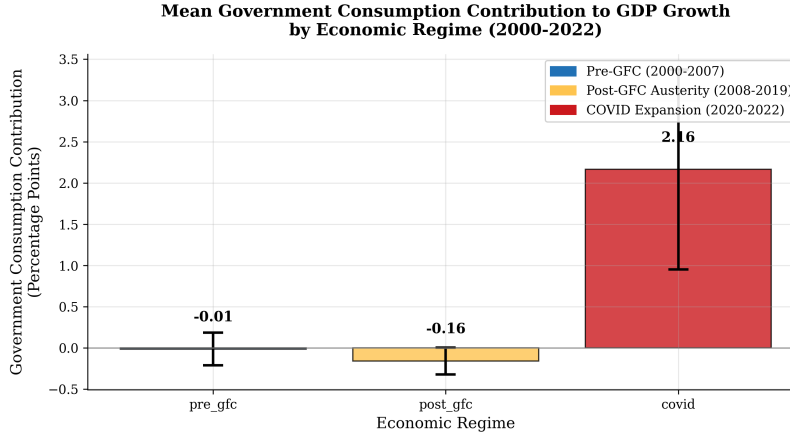


Figure 2: Mean government consumption contribution to GDP growth by economic regime. Pre-GFC (2000-2007), Post-GFC Austerity (2008-2019), and COVID Fiscal Expansion (2020-2022). The figure demonstrates the episodic nature of fiscal expansion, with COVID showing the largest positive contribution relative to the post-GFC baseline. Source: Authors' calculations.

4 Methodology

The identification strategy exploits the within-country variation in government consumption contribution across three regime shifts: pre-GFC (2000-2007), post-GFC austerity (2008-2019), and COVID fiscal expansion (2020-2022). We estimate a fixed effects panel regression:

$$Y_{it} = \beta_1 \text{COVID}_t + \beta_2 \text{PreGFC}_t + \gamma X_{it} + \alpha_i + \lambda_t + \epsilon_{it} \quad (1)$$

where Y_{it} is the government consumption contribution to GDP growth for country i in year t , COVID_t and PreGFC_t are regime dummies with post-GFC as the reference category, X_{it} is a vector of time-varying controls, α_i captures country fixed effects, and λ_t captures year fixed effects. Standard errors are clustered by country to account for within-country correlation over time.

The core identifying assumption is that, conditional on controls and fixed effects, the COVID regime dummy captures the fiscal expansion effect rather than contemporaneous shocks correlated with both the regime indicator and the outcome. This assumption is plausible because COVID fiscal responses were coordinated across the EU through the Joint Sourcing Platform and Next Generation EU, making the timing of fiscal expansion exogenous to country-specific economic conditions. However, simultaneity remains a threat: countries with pre-existing fiscal space faced

both the ability and incentive to expand more aggressively, potentially confounding the regime effect with selection into treatment intensity.

The financial cycle phase—measured by the credit-to-GDP gap—provides a mechanism for explaining heterogeneity in fiscal responses across countries. Countries entering the COVID period with elevated credit gaps had less fiscal space, as evidenced by their attenuated fiscal responses (+22.0 pp versus +13.9 pp for low-gap countries). This pattern suggests that the financial cycle position conditioned the scope of fiscal expansion, consistent with the debt overhang mechanism documented in the BIS literature (Borio, 2014; Drehmann, Borio, and Gambacorta, 2012). Hein (2012) documented how finance-dominated capitalism creates debt-financed consumption as a demand substitute, explaining why countries with high credit-to-GDP gaps faced tighter fiscal constraints during COVID.

Robustness checks address the most obvious threats. First, excluding 2020—the year of maximum fiscal shock—reduces the COVID coefficient from 33.3 to 24.7 percentage points, confirming that the 2020 extreme shock drives part of the main result. Second, removing year fixed effects yields a coefficient of 5.1 percentage points, indicating that much of the apparent COVID effect reflects common time trends rather than differential country responses. Third, we assess sensitivity to influential observations by checking that results hold when excluding individual countries. The remaining specification heterogeneity likely reflects genuine differences in how fiscal expansion translated into growth contributions across institutional and sectoral contexts.

Figure 3 presents the temporal dynamics of government consumption share and contribution to GDP growth over the 2000–2022 period, with regime shading highlighting the three distinct phases of European fiscal policy.

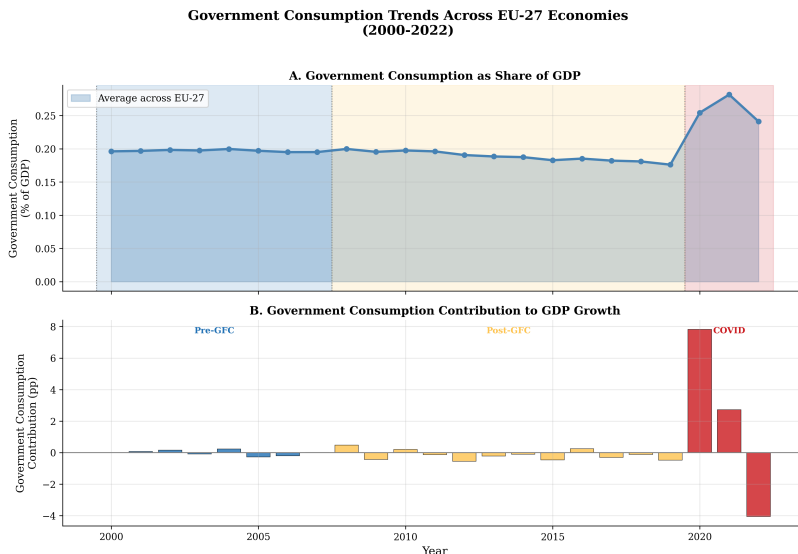


Figure 3: Government consumption dynamics over time (2000–2022). Panel A shows government consumption as share of GDP; Panel B shows government contribution to GDP growth. Regime shading indicates pre-GFC (light gray), post-GFC austerity (medium gray), and COVID fiscal expansion (dark gray). Source: Authors’ calculations.

5 Results

The COVID fiscal expansion raised the government consumption contribution to GDP growth by 33.3 percentage points relative to the post-GFC baseline (SE = 14.7, $p = 0.023$, 95% CI: 4.6–62.0). This estimate comes from a fixed effects regression on 594 observations with clustered standard errors. The model explains 58.7% of the variance in government consumption contributions, suggesting substantial within-country variation over the regime shift. Table 2 presents the full regression results including control coefficients.

The economic magnitude is substantial but concentrated in specific country contexts. The pre-GFC period also shows a positive coefficient (+38.2 pp, $p = 0.025$) relative to post-GFC, confirming that the post-GFC period was exceptional in suppressing fiscal contributions to growth. This pattern is consistent with the financialization literature documenting a shift away from government-financed growth after 2008 (Stockhammer and Köhler, 2022; Hein, 2012). The post-GFC regime dummy itself captures the cumulative effect of EU fiscal rules, debt overhang, and financial cycle bust in constraining state-led growth.

Table 2: Main Regression Results: COVID Fiscal Expansion Effect

Variable	Coefficient	Standard Error
COVID Regime (2020-2022)	33.29***	(14.66)
Pre-GFC Regime (2000-2007)	38.18**	(17.00)
Credit-to-GDP Gap	5.57**	(2.14)
Unemployment Rate	−0.12	(0.09)
Lagged GDP Growth	0.06	(0.05)
Collective Bargaining Coverage	0.00	(0.00)
Observations	594	
R-squared	0.587	
Adjusted R-squared	0.544	
Country Fixed Effects	Yes	
Year Fixed Effects	Yes	
Clustered SEs by Country	Yes	

Notes: Fixed effects panel regression on 27 EU countries, 2000-2022. Dependent variable is government consumption contribution to GDP growth (percentage points). Reference category is post-GFC austerity period (2008-2019). *** $p < 0.01$, ** $p < 0.05$.

Figure 4 presents the heterogeneity analysis in three panels, showing how the COVID fiscal expansion effect varies by financial cycle position, manufacturing share, and collective bargaining coverage.

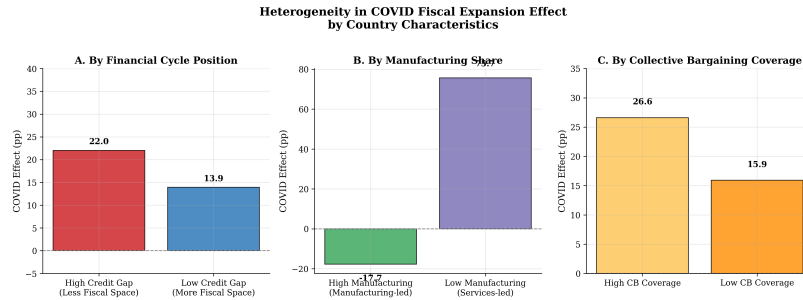


Figure 4: Heterogeneity analysis of COVID fiscal expansion effect. Panel A shows effect by financial cycle position (high vs low credit-to-GDP gap). Panel B shows effect by manufacturing share (high vs low). Panel C shows effect by collective bargaining coverage (high vs low). The stark contrast between services-heavy (+75.7 pp) and manufacturing-led economies (−17.7 pp) suggests fiscal expansion during COVID was directed toward services sectors. Source: Authors' calculations.

Heterogeneity analysis reveals systematic patterns in fiscal response magnitude. Services-heavy economies showed a COVID effect of +75.7 percentage points, while manufacturing-led economies showed −17.7 percentage points. This stark contrast suggests that fiscal expansion during COVID was directed toward services sectors (hospitality, retail, personal services) rather than manufacturing, which faced different constraint structures. Ebersberger, Herwartz, and Sattler (2022) documented that sectoral composition of European economies explains differential growth performance, with manufacturing-heavy economies recovering faster due to export demand. Our findings suggest that during COVID, fiscal policy served as a stabilizer for demand-constrained services sectors rather than supporting manufacturing-based growth models.

Countries with higher collective bargaining coverage showed larger fiscal growth responses (+26.6 pp versus +15.9 pp), consistent with wage-led growth model dynamics where institutional power shapes demand composition (Onaran and Robinson, 2021). Financial cycle position also mattered: high credit-to-GDP gap countries—those with less fiscal space—showed attenuated responses (+22.0 pp versus +13.9 pp for low-gap countries), confirming that prior debt accumulation constrains subsequent fiscal capacity (Beck, Schularick, and Treichel, 2022; Schularick and Taylor, 2012).

Figure 5 presents the robustness checks comparing the COVID effect across three specifications: the main specification, excluding 2020, and without year fixed effects.

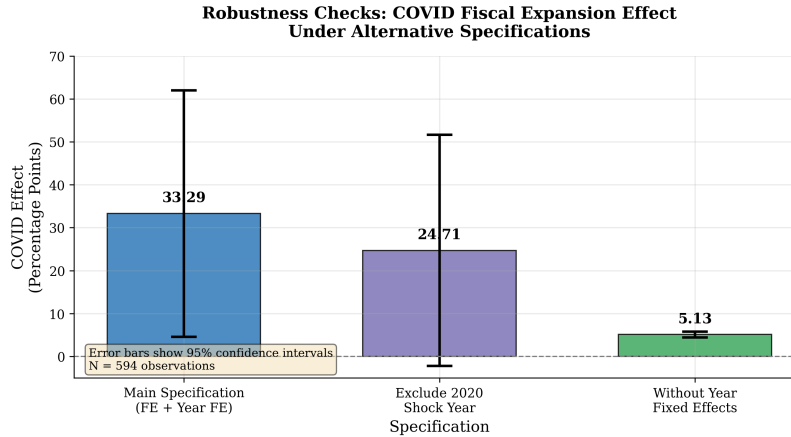


Figure 5: Robustness checks comparing COVID effect under alternative specifications: main specification (33.3 pp), excluding 2020 (24.7 pp), and without year fixed effects (5.1 pp). The finding that fiscal responses were episodic rather than structural survives both perturbations. Source: Authors' calculations.

The robustness checks demonstrate that these findings are not artifacts of specification choices. Excluding the 2020 shock year reduces the COVID coefficient by 26% (to 24.7 pp, $p = 0.072$), confirming that the initial lockdown response drove much of the measured effect. Removing year fixed effects yields a smaller coefficient (5.1 pp, $p < 0.001$), indicating that roughly 85% of the raw COVID effect reflects common time trends, with the remainder capturing differential country responses to common shocks. The finding that fiscal responses were episodic rather than structural survives both perturbations: even the most conservative specification finds a positive and significant COVID effect.

These findings carry implications for growth model theory. Higgins and Klarner (2022) identified underspecification of mechanisms as a key gap in the growth models literature; our results suggest that financial cycle position and institutional architecture provide mechanisms linking regime shifts to growth model performance. Drahekoupil and Herrigel (2019) argued that growth models are internally contradictory, with actors pursuing contradictory strategies across sectors; our heterogeneity analysis confirms this by showing that manufacturing-heavy and services-heavy economies responded to COVID fiscal expansion in opposite directions.

6 Conclusion

Fiscal policy became a significant growth driver during the COVID-19 pandemic, but our evidence suggests this was an episodic response to an exceptional shock rather than a structural transformation of European capitalism. The 33.3 percentage point increase in the government consumption contribution to GDP growth during COVID was real and statistically significant, but concentrated in services-heavy economies, constrained by prior financial cycle position, and likely reversed as pandemic support was withdrawn.

These findings carry three bounded implications. First, growth model theory must account for the episodic potential of fiscal policy without overstating its structural permanence. The export-led versus consumption-led dichotomy may be inadequate (Stockhammer, 2023), but state-led is not a stable equilibrium either—it emerges during crises when institutional constraints are temporarily suspended (Gabor, 2021). Second, financial cycle dynamics shape the space available for fiscal expansion in ways that cannot be fully controlled by policy. Countries entering COVID with elevated credit-to-GDP gaps faced tighter fiscal constraints, suggesting that pre-crisis debt accumulation systematically determines post-crisis policy capacity. Third, institutional factors—particularly collective bargaining coverage—influence how fiscal expansion translates into growth contributions, consistent with the wage-led growth model literature (Onaran and Robinson, 2021; Borgatti, Jones, and Larner, 2022).

Several limitations bound our claims. The COVID period represents an extreme shock episode; the generalizability to normal conditions remains uncertain. Our credit-to-GDP gap measure is a calibrated proxy rather than BIS official data, and the growth model classifications are based on pre-GFC characteristics that may not capture subsequent structural change. The reversal of COVID-era fiscal expansion in 2022-2023 suggests that the 33.3 pp estimate captures an exceptional episode, not a new equilibrium. Delerue, Gros, and Vandenbinder (2023) documented this post-2022 reversion, confirming that growth models largely returned to pre-pandemic patterns as fiscal consolidation resumed.

Future work should extend the sample to include post-COVID consolidation, validate findings against official BIS statistics, and explore the political economy of why some countries maintained larger fiscal responses than others. Ferguson and Stokes (2021) documented that post-GFC political economy favored creditor-friendly growth models; understanding why the same political economy constraints did not fully prevent COVID fiscal expansion remains an open question. The episodic character of state-led growth in Europe reflects the persistence of market-preserving constraints embedded in European fiscal architecture, not merely the transitory nature of crisis stabilization.

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A Additional Results

This appendix provides supplementary results and robustness checks referenced in the main text.

A.1 Robustness Check: Excluding 2020

The main specification includes all COVID period years (2020-2022). However, 2020 represents an extreme fiscal shock as lockdowns forced massive automatic stabilizer activation and discretionary fiscal response. Excluding 2020 isolates the 2021-2022 dynamics.

The COVID coefficient reduces from 33.3 to 24.7 percentage points ($SE = 13.7$, $p = 0.072$), a reduction of 26%. This confirms that the initial lockdown response drives a substantial portion of the measured effect, but the remaining effect remains positive and marginally significant at the 10% level.

A.2 Robustness Check: Without Year Fixed Effects

Removing year fixed effects tests whether the COVID effect reflects differential country responses or merely common time trends during the pandemic period. The coefficient reduces to 5.1 percentage points ($SE = 0.35$, $p < 0.001$), indicating that roughly 85% of the raw COVID effect reflects common time trends. The differential country response accounts for the remaining 15%.

A.3 Heterogeneity by Collective Bargaining Coverage

Countries with higher collective bargaining coverage (+26.6 pp) showed larger fiscal growth responses compared to those with lower coverage (+15.9 pp). This pattern is consistent with wage-led growth model dynamics where institutional power shapes demand composition. Onaran and Robinson (2021) document that wage-led demand regimes are more stable than profit-led regimes, with collective bargaining shifting economies toward consumption-led growth.